

Experiment -2

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Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1: INSTRUCTION PAGE

Chunking Analysis of the Instruction Page

Chunking is a cognitive strategy that involves organizing information into smaller, meaningful units to reduce cognitive load and improve comprehension. The instruction page of the Visual Memory Experiment effectively applies this principle in the following ways:

1. Clear and Sequential Task Flow

- The process is divided into three clearly labeled steps: **Memorize, Recall, and Results**.
- Each step communicates a single goal, helping users mentally prepare for the task without feeling overwhelmed.

2. Logical Grouping of Information

- **Step 1 (Memorize):** Explains the visual exposure phase.
- **Step 2 (Recall):** Describes the memory retrieval activity.
- **Step 3 (Results):** Informs users about performance evaluation.
- A separate highlighted note explains the use of **two presentation styles (chunked and unchunked)**, reinforcing experimental intent.

3. Visual Hierarchy and Layout

- A large, bold title ("Visual Memory Experiment") immediately communicates the purpose.
- Icons next to each step act as visual anchors, improving scanability.
- Adequate spacing between sections prevents visual clutter.
- The **"Begin Experiment"** button is visually prominent, guiding users toward the next action.

4. Reduction of Cognitive Load

- Instructions are concise and written in simple language.
- No unnecessary decorative elements distract users from understanding the task.

Cognitive & UX Benefit

- By chunking instructions into structured steps, users can easily understand the experiment flow, reducing confusion and ensuring fair participation.



Visual Memory Experiment

Exploring the effect of chunking on memory recall



Step 1: Memorize

You will be shown a sequence of icons for a limited time. Pay close attention and try to remember them.



Step 2: Recall

After a brief pause, you'll be asked to select the icons you remember from a larger set.



Step 3: Results

See how well you did and discover how chunking affects memory performance.

Note: You'll experience two different presentation styles - one with grouped (chunked) items and one with ungrouped items. This helps us understand how organization affects memory.

Begin Experiment

FRAME 2: CHUNKING / MEMORIZATION PHASE

Analysis of the Visual Memory Encoding Screen

This screen represents the **visual memory encoding phase**, where users observe and attempt to store visual elements in short-term memory.

1. Purpose of the Screen

- To allow users to **encode visual information** within a limited time.
- To observe how users handle memory load when icons are presented together without explicit grouping (unchunked condition).

2. Key Elements and UI Components

Time Indicator

- A visible countdown timer shows the remaining memorization time (e.g., 7 seconds).
- This creates urgency and simulates real-world memory constraints.

🔗 *Grid of Icons*

- Multiple icons (phone, lightning, home, envelope, bell, bicycle, moon, car, star, camera, key, cup, etc.) are displayed together.
- Icons are visually distinct but presented simultaneously, increasing cognitive demand.

🔗 *Minimal Distraction Design*

- Neutral background and consistent icon styling keep attention focused on the task.
- No interaction is allowed during this phase, preventing rehearsal or correction.

3. How the Memorization Phase Works

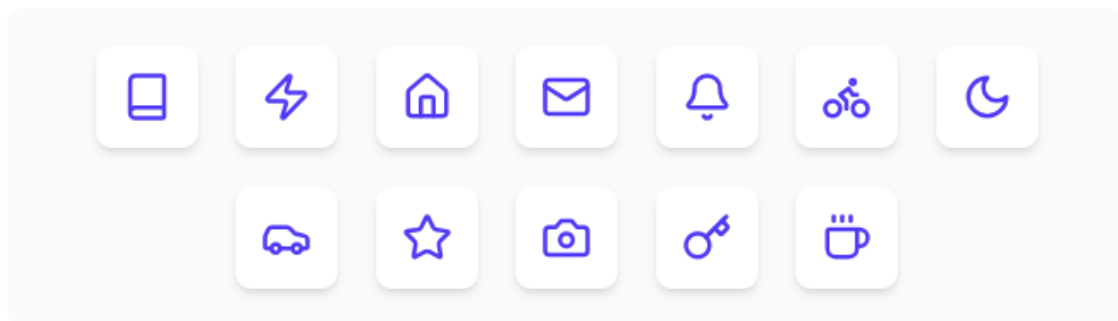
1. Users visually scan the grid of icons.
2. They attempt to form **mental associations or chunks** under time pressure.
3. Once time expires, the screen transitions automatically to the recall phase.

Cognitive & UX Benefits

- Tests **working memory capacity** under load.
- Encourages spontaneous chunking strategies by users.
- Establishes a baseline for comparing chunked vs unchunked performance.

Memorize These Icons

Try to remember all the icons



Time remaining

7s

FRAME 3: RECALL PHASE

Analysis of the Memory Recall (Selection) Screen

This screen represents the **memory retrieval phase**, where users identify icons they remember seeing earlier.

1. Purpose of the Screen

- To measure how accurately users can retrieve stored visual information.
- To evaluate the effectiveness of prior memorization under unchunked conditions.

2. Key Elements and UI Components

Title and Instructions

- The heading clearly states: **“Select the Icons You Remember”**.
- A short instruction reinforces the task goal.

Grid of Selectable Icons

- A larger set of icons is shown, including:
 - Icons from the memorization phase
 - New distractor icons not shown earlier
- Selected icons are visually highlighted, providing immediate feedback.

Selection Counter

- Displays how many icons the user has selected.
- Helps users self-monitor over-selection or guessing behavior.

Action Buttons

- **Clear Selection:** Allows correction before submission.
- **Submit Answers:** Finalizes the recall attempt.

3. How the Recall Phase Works

1. Users compare the displayed icons with their memory.
2. They select icons they believe were previously shown.
3. Distractors test false memory and recognition errors.
4. Submission triggers performance evaluation.

Cognitive & UX Benefits

- Differentiates between **true recall and false recognition**.
- Encourages careful decision-making rather than random guessing.
- Clean interaction design avoids UI-induced errors.

Select the Icons You Remember
Click on the icons you saw in the previous screen
Selected: 5 icons

				
		 ✓		 ✓
				
 ✓		 ✓		 ✓
				

Clear Selection Submit Answers

FRAME 4: RESULT PAGE

Analysis of the Result & Feedback Screen

This screen provides **quantitative feedback** on the user's memory performance.

1. Purpose of the Screen

- To clearly communicate recall performance.
- To explain the effect of presentation style on memory accuracy.

2. Key Metrics Displayed

/// Recall Rate

- Shows the percentage of correctly recalled icons (e.g., 8%).
- Indicates overall memory retention.

Precision

- Displays the accuracy of selected icons (e.g., 20%).
- Highlights the ratio of correct selections to total selections.

Errors

- Shows the number of incorrect selections.
- Reflects false recognition caused by cognitive overload.

3. Result Breakdown

- **Correctly Recalled Icons:** Icons accurately remembered.
- **Missed Icons:** Icons shown earlier but not selected.
- **Incorrect Selections:** Icons selected but never shown.

4. Chunking Effect Explanation

- The system explicitly states that unchunked (random) presentation increases cognitive effort.
- This condition acts as a **control baseline** to compare future chunked presentations.

5. Cognitive & UX Benefits

- Immediate feedback improves self-awareness of memory performance.
- Clear categorization of mistakes helps users understand *why* they performed poorly or well.
- Encourages repeated trials and learning.



Your Results

Presentation Mode: **Unchunked (Random)**

✓ Recall Rate

8%

1 of 12 correct

↗ Precision

20%

Accuracy of selections

✗ Errors

4

Incorrect selections

✓ Correctly Recalled (1)



✗ Missed Icons (11)



✗ Incorrect Selections (4)



 **Chunking Effect**

Unchunked presentation shows items randomly, which requires more effort to encode and recall. This baseline helps us measure the benefit of chunking.

↺ Try Again

PROTOTYPE LINK:

<https://www.figma.com/make/IX6WTnG8bZvUzvdY1HXyS/Memory-Recall-UI-Design?t=qOBGI2AoINTk1i7z-0>